Public Health England

Protecting and improving the nation's health

McCance and Widdowson's The Composition of Foods Integrated Dataset 2021

User guide

Contents

Authors' acknowledgements 3
Introduction 4
Sources of data and methods of evaluation 4
Data formats 6
Details of nutrient data 9
Factors worksheet 9
Proximates worksheet 10
Inorganics worksheet 12
Vitamins worksheet 12
Vitamin fractions worksheet 14
Food labelling 15
'Front of pack' nutrition labelling 16
Tolerances for nutrient values declared on a label 17
Appendices 18
Appendix A: Main data references 18
Appendix B: Food sub-group codes 23
Appendix C: Acronyms, descriptions and units 27
References 36

Authors' acknowledgements

We would like to thank colleagues who worked on previous versions of CoFID, which this version builds on, and/or provided advice on the 2021 dataset.

Colleagues at Quadram Institute Bioscience

Mark Roe and Rachel Berry.

Colleagues at Public Health England

Sakhi Dodhia, Melanie Farron-Wilson, Joanne McArdle, Danielle Weiner, Gillian Swan and Mark Bush.

Colleagues at Department of Health and Social Care

Hannah Koffman and Debby Webb.

Introduction

Public Health England (PHE) is responsible for maintaining data on the nutrient content of the UK food supply to support the National Diet and Nutrition Survey, and funds nutrient analysis of foods commonly consumed in the UK. Data from this work, and complementary data from other sources are published as McCance and Widdowson's 'The Composition of Foods' – the UK food composition tables. Previous data have been published in book form as a series of supplements, each containing extensive data on a specific food group, and also as the summary editions.

The Composition of Foods Integrated Dataset (CoFID) was first published in 2008 and brought together for the first time all the available data in electronic format as a single, consolidated dataset. The 2015 version was an update of the original published in 2008 and included all the new and recently reviewed data available at that point. Foods calculated by recipe or calculated based on another food (for example, foods weighed with waste) were recalculated to include the most recent data for ingredients. A subset of the 2015 data were also published in book form as McCance and Widdowson's The Composition of Foods seventh summary edition. A further update of CoFID in 2019 included data from a 2015 fruit and vegetable nutrient analysis project and new and revised data on whole and toasted almonds. This updated version (2021) of CoFID incorporates data from a 2020 report on nutrient analysis of key cuts of pork including updates for associated foods and recipes within the pork section. Data on the following foods have been corrected: vitamin C in watercress; iodine in soya yogurt; and vitamin E, sugars and biotin in boiled and steamed sweet potato. All other values remain unchanged from CoFID (2019).

Sources of data and methods of evaluation

It is essential that food composition data are regularly updated for a number of reasons. New fresh, ambient, frozen, and processed foods are introduced and the nutrient content of existing foods change. This can happen when there are new varieties or new sources of supply for the raw materials. New farming practices can affect the nutritional value of both plant and animal products. New manufacturing practices, including changes in the type and amounts of ingredients, and changes in fortification practice can affect the content of processed foods. Many foods have been reformulated in line with government public health initiatives, including reductions in the amount of fat, trans fatty acids, saturated fat, sugar and salt added. Methods of preparation and cooking in the home also change.

The analytical survey reports from 2011 onwards are available to download in electronic form.

Earlier reports can be requested from Food Databanks National Cability (FDNC). These reports comprise raw laboratory data and have not been evaluated to the same extent as data incorporated into CoFID.

The Composition of Foods series of publications in book form are listed below:

Cereals and Cereal Products, supplement (1988)
Milk Products and Eggs, supplement (1989)
Vegetables, Herbs and Spices, supplement (1991)
Fifth Summary Edition (1991)
Fruit and Nuts, supplement (1992)
Vegetable Dishes, supplement (1992)
Fish and Fish Products, supplement (1993)
Miscellaneous Foods, supplement (1994)
Meat, Poultry and Game, supplement (1995)
Meat Products and Dishes, supplement (1996)
Fatty Acids, supplement (1998)
Sixth Summary Edition (2002)
Seventh Summary Edition (2015)

Many of the values included in CoFID have been taken from the seventh summary edition and detailed supplements, themselves mainly derived from the analytical surveys programme. All the data in the supplements were included in the original version of CoFID published in 2008. The data were extensively revised and updated for the 2015 revision of CoFID some further revisions were made for the 2019 version, mainly to fruit and vegetables. Additional revisions were made to incorporate data on pork in the latest 2021 dataset The main data source for each food is provided, where known, in the 'main data reference' column and refers to the references provided in Appendix A, which lists the reports and scientific literature from which data were taken.

Some foods that were included in the 2008 CoFID have not been reviewed or validated in the 2015, 2019 or 2021 update and those foods are published separately in a file containing data for 'old' foods. This data can be used in addition to the 2021 dataset, but users should be aware that the values for some nutrients may not represent those foods as currently consumed.

Where the values in the tables were derived by direct analysis of the foods, care was taken when designing sampling protocols to ensure that the foods analysed were representative of those consumed by the UK population. For most foods a number of samples were purchased at different shops, supermarkets or other retail outlets, and, where appropriate, foodservice outlets or catering suppliers. Samples analysed were composite samples, consisting of equal quantities of each sub-sample purchased.

When the composite sample was made up from a number of different brands of food, the proportion of the individual brands purchased was related to their relative shares of the retail market. Full details of samples are available in the reports in the 'main data references' column (Appendix A). If the food required preparation prior to analysis, techniques such as washing, soaking, cooking and so on, were as similar as possible to normal domestic practices. Cooking methods were based on manufacturer's recommendations for pre-packaged foods and methods for non pre-packed foods were based on review of 'usual' consumer preparation. Details of preparation procedures are available in the reports in the main data reference column (Appendix A).

Where data from literature sources were included, preference was given where the food was similar to that in the UK, where full details of the sample, method of preparation and analysis were given, and where the results were presented in a detailed and acceptable form. EuroFIR (European Food Information Resource) datasets and tools were used to help evaluate data values, and to calculate values for foods where suitable analytical or literature data were not available.

Where processed foods with brand names are included, they are restricted to leading brands with an established composition. No inference should be drawn from the inclusion of data for a particular brand.

The final selection of values published is dependent on the judgement of the compilers and their interpretation of the available data. Due to the large natural variability of foods, it is unlikely that a particular item will have precisely the same composition as given in these tables. This is particularly true for unprocessed foods such as cereals, dairy products, eggs, meat, fish, fruit and vegetables. The values published should be regarded as typical rather than definitive for the foods described.

Users are advised to consult other sources of data (for example, product labels, manufacturers' data, published analytical reports) where appropriate, depending on their particular needs or interests, for the food item under consideration. It should be noted that manufacturers can and do change or reformulate their products and this will influence nutrient content. This is particularly relevant for foods where nutrients are added for fortification purposes, or for technological purposes, such as antioxidants or colouring agents. Information on processed foods, including fortification levels and reformulations, is often available from manufacturers' websites and from retailers.

Data formats

CoFID data are provided as Excel workbooks containing data in worksheets related to nutrient groups.

Format of Excel files

The Excel workbook consists of the following worksheets:

List of worksheets in CoFID 2021
Factors
Proximates
Inorganics
Vitamins
Vitamin Fractions
Saturated fatty acids per 100g fatty acids
Saturated fatty acids per 100g food
Monounsaturated fatty acids per 100g fatty acids
Monounsaturated fatty acids per 100g food
Polyunsaturated fatty acids per 100g fatty acids
Polyunsaturated fatty acids per 100g food
Phytosterols
Organic acids

Each sheet contains column headings in rows 1 to 3 of the spreadsheet, then data values for each individual foodstuff such that data values for a specific food code will occur in the same row in each of the worksheets.

The following notes apply to the data values:

Nutrient values are expressed per 100g of the food except in the case of alcoholic beverages which are presented per 100ml.

A trace value for a nutrient is represented by Tr.

Where a nutrient is present in significant quantities, but there is no reliable information on the amount, the value is represented by N.

In the Excel files, it is not possible to append nutrient-specific footnotes. However, these are being updated in the Composition of Foods Integrated Dataset searchable website. Food-specific footnotes do appear in the Excel version.

Food identification

Food code

Food code is a number, up to 6 digits, representing the unique CoFID code representing each food. The 2-digit prefix generally refers to the food codes used within the book supplements relating to specific food groups, for example, 11- for cereals and cereal

products, 12- for milk products and eggs, 16- for fish and fish products. For foods where new data have been incorporated into an existing food, a new food code has been given using the prefix for the appropriate supplement. The food codes should not be considered to have any particular significance as the electronic datasets do not relate exactly to book publications, and each food has a unique food code.

Food Name (NAME)

The food name has been chosen as that most recognisable and descriptive of the food referenced.

Description (DESC) Information

Description (DESC) Information given under the description describes the nature of the samples taken for analysis. Sources of values derived, either from the literature or by calculation, are also indicated under this heading.

Group (GROUP)

In these files (but not in the printed publications) a one, 2 or 3 letter code is assigned to every food. The code letter(s) provide identification of the food group and food type to which the food belongs. A full list of the codes and their description is given in Appendix B.

Previous (PREV)

Assigned to each food which has an earlier food code with different nutrient values associated with it. It is a number, up to 6 digits, indicating previous food codes (4th, 5th or 6th edition or a supplement). Some foods may have more than one previous code associated with them.

Main data references (COMMENTS)

The main data reference indicates the principal report(s) or publication(s) from which the majority of the data for the food code are taken. Values for individual nutrients within each code may be taken from different sources, calculated or estimated from other codes. For foods that do not have an analytical report or literature source that can be referred to as the main data reference, the food description should indicate how the data have been estimated (for example, from manufacturer's data, calculated from related codes or calculated as a recipe). In some cases, there is a main data reference referring to analytical data and the description indicates that industry data have also been used to update some nutrients, usually sodium, sugars, fats or added minerals and vitamins.

Details of nutrient data

For a more detailed definition and expression of the nutrients refer to the introductory pages of the seventh summary edition of McCance and Widdowson's The Composition of Foods (1) and of the supplement publications.

The more significant points for certain nutrients are provided in the notes below for convenience. For some nutrients, data which is 'old' and were analysed significantly earlier and on a different sample to the bulk of the nutrients are available. These data have not been updated but are the only data available and may be of interest to some users and are available as a separate file. This applies to fibre fractions, fibre analysed by the Southgate method, and sulphur.

Factors worksheet

Edible conversion factor (EDPOR)

Many foods are purchased or served with material that is clearly inedible or material that might be discarded as inedible by some consumers. For the purposes of this dataset 'waste' encompasses both types of material, which might include, for example:

- outer leaves or stalks of vegetables
- stones, pips or peel of fruit
- nut shells
- fish skin and bone
- meat fat and bones
- liquid content of canned foods

The edible conversion factor allows calculation of the nutrient content of foods when the inedible material is included in the weight and refers to the proportion of edible material remaining after the waste has been removed. The factor will vary between different samples of the same food and these values should be treated as a guide to the typical proportion of inedible waste. In the dataset foods containing inedible portions always have 'weighed with' in their name.

Specific gravity (SPECGRAV)

Specific gravity is the ratio of the density (mass of a unit volume) of a food to the density (mass of the same unit volume) of water.

Nitrogen conversion factor (NCF)

Nitrogen conversion factor is the factor used to calculate protein from total nitrogen. See Protein.

Glycerol conversion factor (GCF)

Glycerol conversion factor is used to allow the calculation of the total fatty acids in a given weight of food. See Fatty acids.

Proximates worksheet

Water (WATER)

For most foods, water has been analysed using gravimetric methods. In some cases where protein, fat or carbohydrate have been updated based on industry data, the water value has been estimated by calculation (100 – (protein + fat + available carbohydrate + dietary fibre + ash)).

Protein (PROT)

For most foods, protein is calculated by multiplying total nitrogen values (TOTNIT) by the factors provided in the 'Nitrogen conversion factor' column of the 'Factors' worksheet, as described in the introduction of McCance and Widdowson's The Composition of Foods seventh summary edition. Unless stated otherwise, a factor of 6.25 is used based on the assumption that proteins contain 16% nitrogen. The proportion of non-protein nitrogen is high in many foods, notably fish, fruits and vegetables. In most of these, however, this is amino acid in nature and therefore little error is involved in the use of a factor applied to the total nitrogen, although protein in the strictest sense is overestimated. For those foods which contain a measurable amount of non-protein nitrogen in the form of urea, purines and pyrimidines (for example, mushrooms) the non-protein nitrogen has been subtracted before multiplication by the appropriate factor.

Fat (FAT)

The fat in most foods is a mixture of triglycerides, phospholipids, sterols and related compounds. The values in the tables refer to total fat and not just to triglycerides.

Carbohydrate (CHO)

Total carbohydrate and its components, starch, total and individual sugars (glucose, galactose, fructose, sucrose, maltose, lactose) and oligosaccharides, but not fibre, are wherever possible expressed as their monosaccharide equivalent. The values for total carbohydrate in the dataset have generally been obtained from the sum of analysed values for these components of 'available carbohydrate', contrasting with figures for carbohydrate 'by difference', which are sometimes used in other food tables or on the labels of processed foods. Such figures are obtained by subtracting the measured weights of the other proximates from the total weight and many include the contribution from any dietary fibre present, as well as errors from the other analyses.

Energy value (KCALS)

Calculated using the conversion factors: protein 4 kcal/g, fat 9 kcal/g, carbohydrate (available, expressed as monosaccharides) 3.75 kcal/g and alcohol 7 kcal/g.

Energy value (KJ)